

Transparent, hydrophobic, self-polishing organosilicon-modified zinc acrylate polymer coating for marine antifouling

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ABSTRACT

Considering the increasing efforts toward marine environment protection, the replacement of the conventional self-polishing antifouling coatings with their ecofriendly alternatives is urgently required. The key aspect of this transformation is the augmentation of the antifouling capabilities of matrix resins, which enables a substantial reduction in the use of antifouling agents. Consequently, research on high-performance self-polishing antifouling matrix resins is of utmost importance. Herein, to prepare a multifunctional zinc acrylate polymer for marine antifouling, an acrylate siloxane monomer (ASM) that undergoes free-radical polymerization with acrylate monomers was synthesized by grafting a hydroxyl-containing organosilicon intermediate (RSN-6018) with hydroxyethyl methacrylate. The prepared coating demonstrates excellent transparency in the short term after the addition of the ASM, addressing the issue of the poor compatibility of the organosilicon-modified zinc acrylate polymer. After immersion in seawater for 35 d, the organosilicon-modified zinc acrylate polymer exhibits a stable hydrolysis performance, facilitating the self-polishing of the coating. Because of the presence of the siloxane structure, the coating maintains stable hydrophobicity even after soaking in seawater for 35 d, demonstrating a contact angle of $115.0^\circ \pm 1^\circ$. The organosilicon-modified zinc acrylate polymer coating has an inhibition rate of up to 99 % against *Staphylococcus aureus* and exhibits antifouling effects lasting for more than two months in the actual marine environment. Overall, the proposed method offers a high level of innovation and provides an effective and novel approach for preparing modified self-polishing antifouling resins.

1. Introduction

Marine biological fouling refers to the settlement and accumulation of microorganisms, plants, and animals in seawater on submerged surfaces [1]. Biofouling increases drag, which results in high fuel consumption, greenhouse gas emissions, and corrosion, ultimately reducing the operating life of ocean equipment [2,3]. Using antifouling coatings is the most straightforward method to mitigate the aforementioned issues and reduce the commercial losses [4].

Tin-free self-polishing antifouling coating is a widely used antifouling coating. This type of coating relies on the hydrolysis of water-soluble polymer compounds [5] in the base resin in seawater environments. This chemical reaction considerably alters the surface properties of coatings, which facilitates the release of antifouling agents, achieving the prevention of dynamic and static fouling [6]. Although this method

can effectively reduce biofouling to a certain extent, its negative effects on the environment cannot be disregarded [7]. Therefore, its ecofriendly alternatives are being developed. Wang et al. [8] used the positive charge of quaternary ammonium salts to add the natural antibacterial [9–12] agent eugenol to polyacrylate side chains, enhancing the efficiency of eugenol and reducing the damages on the marine environment. Furthermore, Ma et al. [13] crafted a branched copolymer to precisely control the degradation and release rate of 4,5-dichloro-2-octylisothiazolone to reduce the use of antifouling agents. Lejars et al. [14] synthesized a p-MATM2-b-p(MMA-co-BMA) diblock copolymer via a reversible addition–fragmentation chain-transfer process. Coatings comprising this copolymer exhibited evolving surface properties in artificial seawater, initially demonstrating hydrophobicity, and displayed an antifouling effect lasting for 16 months in real seawater. These studies support the development of highly sustainable and effective self-

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polishing antifouling coatings.

Recently, hydrophobic paints have attracted widespread attention and can improve the fouling-release performance of coatings [4,15,16] because of their large contact angles [17,18] and low elastic moduli [19,20] via the introduction of hydrophobic groups [21,22], such as long-chain hydrocarbons [23–26], fluorinated organic compounds [27,28], and organosilicon compounds [29–31], reducing the consumption of antifouling agents. In many of the hydrophobic groups mentioned above, the siloxane main chain of organosilicon compounds can exhibit excellent rotational flexibility [32], which considerably improves the hydrophobicity of coatings, and is widely employed for enhancing the hydrophobic properties of materials [33–36]. Therefore, to improve the hydrophobic performance of zinc acrylate polymers, the introduction of an organosilicon structure into the self-polishing zinc acrylate coating for its hydrophobic modification is an effective method for enhancing the antifouling performance.

In this study, we synthesize an acrylated siloxane monomer (ASM) that undergoes free-radical copolymerization with acrylate monomers to prepare a zinc acrylate polymer coating that integrates transparency, hydrophobicity, and self-polishing properties, realizing the organosilicon modification of the zinc acrylate polymer. The organosilicon-modified zinc acrylate polymer maintains good transparency even after soaking in seawater for 7 d, indicating its good compatibility. The siloxane structure in the polymer has several hydrophobic properties, and it exhibits a stable hydrophobic performance, demonstrating a contact angle of $115.0^\circ \pm 1^\circ$ even after soaking in seawater for 35 d. A stable self-polishing effect of the coating is achieved via the hydrolysis of the zinc ions of the zinc acrylate polymer in seawater. When the ASM content is 0.5, the coating remains flat at a sailing speed of 16 knots, demonstrating the excellent adhesion strength of the coating during sailing. Owing to the distribution of a large amount of organic acids on the coating surface, the inhibition rate of the coating against *Staphylococcus aureus* (*S. aureus*) is as high as 99 %. Furthermore, the coating exhibits an antifouling period of more than two months in the actual marine environment. In summary, because of its unique structural composition, encompassing ester groups, zinc ions, polysiloxane, and organic acids, the coating exhibits an exceptional marine antifouling performance, as evidenced by the positive results of the antibacterial and antifouling experiments as well as real-world marine antifouling tests. This research offers new perspectives and methodologies for advancing marine coating technology and considerably contributes to the preservation of marine ecosystems and the enhancement of the longevity of marine infrastructures.

2. Experimental

2.1. Materials

An organosilicon intermediate (RSN-6018; Table S1) was purchased from Dow Corning, USA [37]. Stannous octoate was purchased from Sinopharm Chemical Reagent Co., Ltd. (Shanghai, China). Tetraisopropoxide titanate (TPT) and hydroxyethyl methacrylate (HEMA) were procured from Aladdin (Shanghai, China). Toluene was purchased from Xilong Scientific Co., Ltd. (Chengdu, China). Ethyl acrylate (EA), butyl acrylate (BA), acrylic acid (AA), naphthenic acid, xylene, n-butanol, and butyl acetate were purchased from Datang Chemical Group Co., Ltd. (Qingdao, China). Acrylamide (AM) and dodecanethiol were purchased from Sinopharm Chemical Reagent Co., Ltd. (Shanghai, China). Cyclohexyl methacrylate (CHMA) and lauryl acrylate (LA) were acquired from Guangzhou Swan Chemical Co., Ltd. (Guangzhou, China). Azodiisobutyronitrile (AIBN) was purchased from the Tianjin Damao Chemical Reagent Factory. (Tianjin, China). Zinc oxide was purchased from Haishun New Material Co., Ltd. (Zibo, China). Benzoic acid was purchased from Shanghai Yi'en Chemical Technology Co., Ltd. (Shanghai, China). Rosin resins were purchased from Zhiyuan Chemical Products Co., Ltd. (Zhengzhou, China).

2.2. ASM synthesis

First, 105 g of Z-6018 (0.38 mol), 0.5 g of TPT, 0.5 g of stannous octoate, and 60–70 mL of toluene were added to a 500 mL four-necked round-bottom flask fitted with the Dean–Stark assembly. Thereafter, 50 g of HEMA (0.38 mol) was added dropwise to the flask, maintaining the temperature at 80 °C. Afterward, the reaction mixture was heated to 120 °C under stirring until the calculated amount of water was collected through the Dean–Stark apparatus [37]. TPT promotes the desirable condensation between silanol –OH and acrylic –OH and prevents the self-condensation of the organosilicon intermediate. The ASM was dried in an oven at 125 °C and ground into powder for material characterization.

2.3. Preparation of the ASM-modified zinc acrylate resin

ASM_x-Zn was synthesized via free-radical copolymerization and dehydration condensation. The synthesis steps are presented in Fig. 1. First, a monomer mixture (M1) was prepared by mixing different contents of the ASM with 48.3 g BA, 18.9 g EA, 13.7 g AA, 5.1 g AM, 20.3 g LA, and 1.7 g CHMA. Next, 10 wt% of M1, a xylene: n-butanol mixture (xylene: n-butanol = 4:1; 100 wt% of M1), dodecanethiol (0.8 wt% of M1), and AIBN (0.15 wt% of M1) were added to a 500-mL three-necked flask. The reaction was conducted at 85 °C for 15 min. Afterward, 90 wt % of M1 and AIBN (0.65 wt% of M1) were added to the three-necked flask at 85 °C in nine equal portions at 15 min intervals. After intermittent dosing, the reaction was continued for 30 min. Subsequently, AIBN (0.20 wt% of M1) was added to the three-necked flask in two portions at 20 min intervals and then reacted at 85 °C for 2 h to obtain an acrylate prepolymer comprising different ASM contents (ASM_x, x = 0, 0.25, 0.5, 1.0, and 1.5). After cooling to 70 °C, 9.3 g ZnO, 14.7 g rosin resins, 14 g benzoic acid, 11.6 g naphthenic acid, and 49.5 g solvent (xylene: n-butanol: butyl acetate = 4.4:1:10.6) were added to the aforementioned system and reacted for 2 h at 85 °C to obtain a zinc acrylic copolymer with different ASM contents (ASM_x-Zn, x = 0, 0.25, 0.5, 1.0, and 1.5; Table S2).

3. Characterization

3.1. Chemical structure analysis of ASM_x-Zn

3.1.1. Attenuated total reflection (ATR)-infrared (IR) spectroscopy

The ATR-IR spectra were recorded using a German BRUKER TENSOR II transform IR spectrometer. The powder/liquid/film samples were laid flat on a sample stage, and the IR absorption peaks were measured in the 4000–400 cm^{−1} range. The test was performed against an air background.

3.1.2. Thermogravimetry analysis (TGA)

A NETZSCH STA 449 F3 Jupiter® synchronous thermogravimetry–differential scanning calorimetry thermal analyzer was employed to measure the weight loss of the sample at 30 °C–800 °C. An appropriate amount (3–5 mg) of the sample was placed in an alumina crucible, and nitrogen was used as the purge and protection gas. The specific test conditions were a nitrogen gas intake of 50 mL•min^{−1}, heating rate of 10 °C•min^{−1}, and temperature range of 30 °C–800 °C.

3.1.3. Proton nuclear magnetic resonance (¹H NMR) spectroscopy

The ¹H NMR data were obtained utilizing Bruker AVANCE III 600 using dimethyl sulfoxide-*d*₆ as the solvent and tetramethylsilane as the internal standard. Use MestReNova software to deeply analyze the nuclear magnetic resonance (NMR) data, accurately resolve the chemical shift values corresponding to each signal peak, and analyze the structure of ASM_x-Zn.

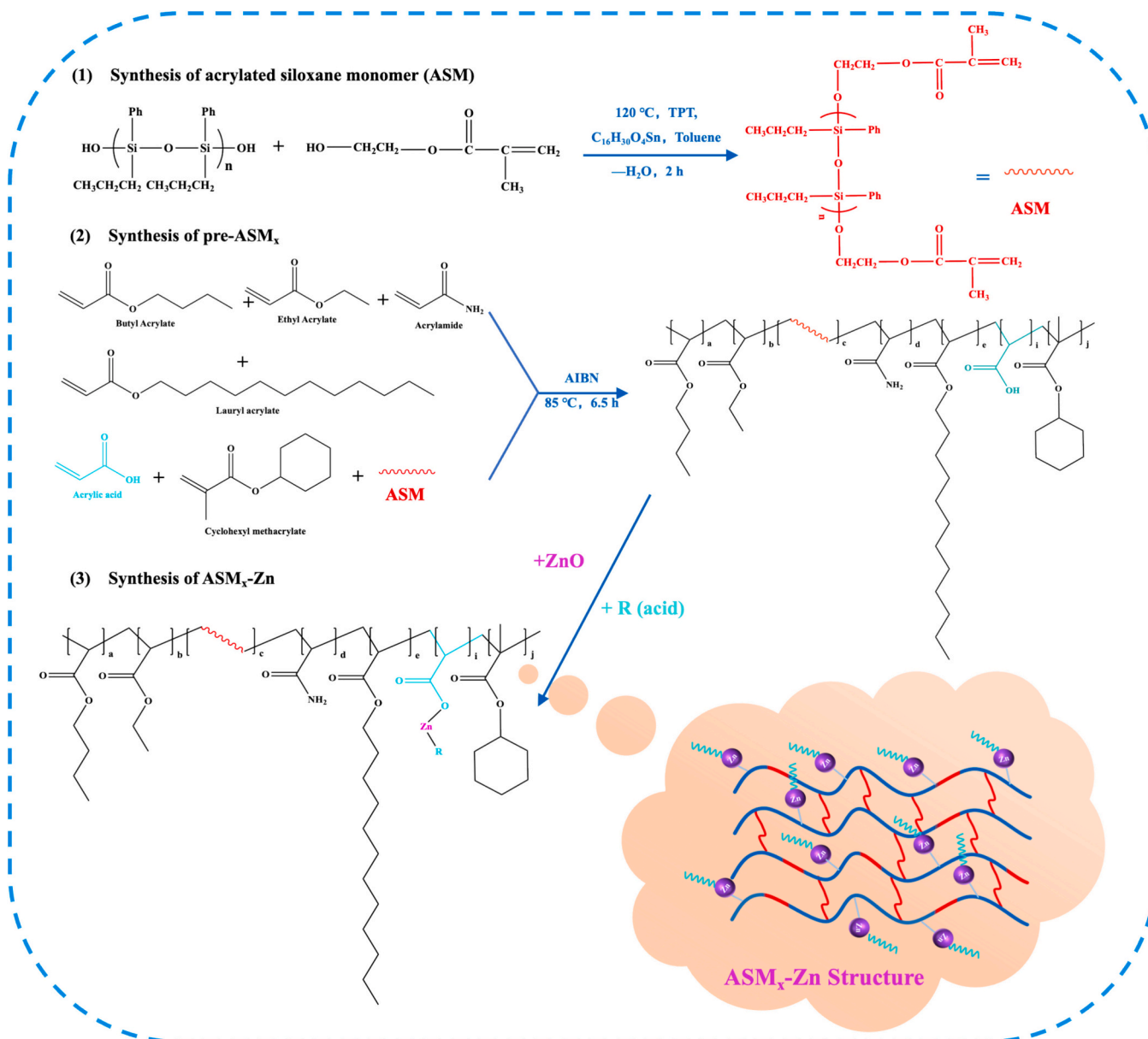


Fig. 1. Synthesis process of the ASM_x-Zn polymer.

3.2. X-ray photoelectron spectroscopy (XPS) measurements of ASM_x-Zn coatings

The ASM_x-Zn polymer was applied to a polytetrafluoroethylene film and dried. Afterward, the film was cut into samples with the dimensions of 5 × 5 mm². ASM_{0.5}-Zn was considered a representative sample. The sample was placed into the sample chamber of a Thermo Scientific K-Alpha XPS instrument. When the pressure in the sample chamber was lower than 2.0 × 10⁻⁷ mbar, the sample was introduced into the analysis chamber. The beam spot size was 400 μm. The working voltage and filament current were 12 kV and 6 mA, respectively. The element distribution on the coating surface was tested under the conditions of a full spectrum scan in an energy range of 150 eV with a step size of 1 eV and a narrow spectrum scan in an energy range of 50 eV with a step size of 0.1 eV.

3.3. Hydrolysis properties

Prior to the immersion experiments, acrylonitrile–butadiene–styrene

(ABS) plates with the dimensions of 8 × 10 cm² were cleaned using ethanol, dried, and weighed (W_0). The ASM_x-Zn resin was applied to the ABS boards, which were subjected to natural evaporation for 48 h and subsequently dried in an oven at 40 °C for 48 h. The weight of each coated ABS board was recorded as W_t . The samples were immersed in real seawater, withdrawn every seven days, rinsed with deionized water, and dried in an oven at 40 °C for 48 h. The mass of the dried samples was recorded as W_i ($i = 1-5$). The percentage of the remaining mass of the coating after weight loss via immersion (W_s) was calculated using the following formula:

$$W_s = \frac{W_i - W_0}{W_t - W_0} \times 100\% \quad (1)$$

3.4. Water contact angle measurements

The contact angles of the polymer coatings were measured using an optical contact angle measuring device (Dataphysics Contact Angle System OCA40, Germany). The polymer was applied to an 8 × 10 cm² ABS plate, which was first naturally dried for 48 h and subsequently

dried at 40 °C for 48 h. The changes in the static contact angle were measured before and after five weeks of immersion in real seawater. Before the test, the surface was blown clean using nitrogen and placed flat on the sample stage. Next, 2 μ L of deionized water was added to the coated surface using an automatic sampling device. The contact angles were recorded using the sessile method at 25 °C. For each sample, multiple sets of tests were conducted at different locations, and the average value of at least three sets of valid data was adopted as the contact angle of the sample.

3.5. Surface morphology and roughness

The polymer was applied to $8 \times 10\text{-cm}^2$ ABS boards, which were dried first naturally for 48 h and subsequently in an oven at 40 °C for 48 h. The sample boards were immersed in real seawater for 14 d. A laser confocal microscope (LEXT OLS4000) was used to observe the microscopic morphology and roughness of the polymer coating surfaces. Furthermore, the changes in the morphology and roughness of the polymer coatings before and after static immersion in real seawater were determined. The arithmetic average roughness (R_a , which reflects the absolute value of roughness over a reference length) was used to quantify the surface roughness of the coatings.

3.6. Dynamic simulation experiment

The polymer was applied to $250 \times 100\text{-mm}^2$ ABS boards with a controlled thickness of 400 μ m and dried first naturally for 48 h and subsequently at 40 °C for 48 h. The sample boards were placed into a dynamic simulated paddling experimental setup (Fig. S1) containing natural seawater at a speed of 80 rpm, and the flatness of the coatings was tested under seawater washout conditions.

3.7. Adhesion test

The adhesion strength of $\text{ASM}_x\text{-Zn}$ coatings to a glass fiber-reinforced epoxy resin substrate was measured using an adhesion tester (PosiTest AT-A Automatic) based on the ASTM Standard D 4541. The pull-test adhesion data were obtained by detaching an aluminum dolly (diameter = 20 mm) at a speed of 0.2 MPa/s. The tests were performed at five different regions of each sample, and the average value was reported [38].

3.8. Pencil hardness test

Pencil hardness tests were performed on the coatings after five weeks of their immersion in seawater based on the ASTM-D3363 test standard—Coating preparation reference hydrolysis performance test section. Each sample was tested three times.

3.9. Flexibility

$\text{ASM}_x\text{-Zn}$ was applied to a $5 \times 5\text{-cm}^2$ polyethylene terephthalate (PET) film at a coating thickness of approximately 100 μ m, and the coated film was subsequently immersed in real seawater for one week. Thereafter, the sample film was removed and naturally dried for 3 d. The dried coating was bent into a U-shaped cross section possessing a diameter of 2.5 cm. Subsequently, the force was released to return the film to its original shape, completing a bending cycle that was repeated 50 times. The results were photographed using a digital camera.

3.10. Transparency test

The visible (vis) spectra of the coatings were recorded using an ultraviolet–vis (UV–vis) spectrophotometer (Persee, TU-1901). All the coatings were coated at a thickness of 150 μ m on $5 \times 5\text{-cm}^2$ PET films, and the statistics associated with the vis spectra were determined using

the PET substrate as the baseline. The test wavelength ranged from 200 to 800 nm.

3.11. Antifouling measurements

3.11.1. Antibacterial test

First, 5 mL of $\text{ASM}_x\text{-Zn}$ was dropped onto an ABS disk possessing a diameter of 3 cm. The disk was spin coated using a spin coater (KW-4 A, SETCAS LLC) at a speed of 3000 rpm for 3 s. Thereafter, it was dried first naturally for 48 h and subsequently at 40 °C using an oven for 48 h. The plate count method was employed to evaluate the effects of the $\text{ASM}_x\text{-Zn}$ coating. *Escherichia coli* (*E. coli*) and *Staphylococcus aureus* (*S. aureus*) were used as the model bacteria, representing gram-positive and gram-negative bacteria, respectively. Initially, a single colony was cultured in a nutrient broth at 37 °C for 24 h to obtain an initial bacterial suspension with a concentration of approximately 10^9 CFU/mL. Thereafter, each sample coating was immersed in 5 mL of the initial bacterial suspension for 24 h at 37 °C, enabling the bacteria to come into contact with the coating and potentially be affected by its antibacterial components. After immersion, the bacterial suspension containing the coated samples was diluted to 10^{-5} using the ten-fold serial dilution method. This step was necessary to decrease the bacterial concentration and facilitate the counting of colonies on the solid media. Subsequently, 100 μ L of the diluted bacterial suspension was evenly spread onto the solid culture media, which were then incubated at 37 °C for 24 h. The colony formation was observed and recorded to compare the antibacterial effects of different sample coatings. To quantify the antibacterial performance of the coating, we calculated the antibacterial rate (A) using the following formula:

$$A = \frac{N_0 - N_i}{N_0} \times 100\% \quad (2)$$

where N_0 and N_i represent the colony counts of the blank control and experimental samples, respectively. A higher antibacterial rate (A) indicates the better antibacterial properties of the coating. Throughout the experiment, all the materials used were sterilized using an autoclave to ensure sterile conditions. Additionally, all the operations were performed in a sterile laminar flow cabinet to prevent external contamination from affecting the experimental results.

3.11.2. Algal inhibition test

The prepared coatings were applied to $5 \times 5\text{-cm}^2$ PET films and dried first naturally for 48 h and subsequently at 40 °C using an oven for 48 h. Because algae are the primary microorganisms in biofilms, two common algae in marine environments, *Chlorella pyrenoidosa* (*C. pyrenoidosa*) and *Nitzschia closterium* (*N. closterium*), were selected as model organisms to evaluate the antifouling performance of the coating. In an RXZ-380C artificial climate chamber, the algae were cultivated in an f/2 medium using a light intensity of 4000 lx with a light/dark cycle of 12 h:12 h at a temperature of 22 ± 1 °C. When the exponential growth phase was reached, the optical density of the algal suspension was diluted to 0.08 (680 nm) using sterilized seawater. The sample coatings were subsequently immersed in 40 mL of the diluted algal suspension, and the concentration of the algal suspension was measured daily at 680 nm over a period of 7 d. The antialgal rate over 7 d was calculated using the following formula:

$$S = \frac{S_0 - S_i}{S_0} \times 100\% \quad (3)$$

where S is the antialgal rate, and S_0 and S_i are the algal suspension concentrations of the control and experimental groups, respectively.

3.12. Marine field tests

The antifouling performance of the polymer coating was tested based on the Chinese standard GB/T 5370–2007 Test Method for Anti-fouling Paint Sample by Shallow Sea Immersion. The offshore polyvinyl chloride (PVC) wallboard frames designed in the laboratory were divided into 14 pieces, and each piece was applied with the polymer coating over an area of $140 \times 160 \text{ mm}^2$ at a thickness of 150–200 μm . Thereafter, they were dried in a ventilated place. The coated frames were immersed in seawater at a depth of 0.5–2.0 m in the Qingdao Middle Harbor (Qingdao, Shandong Province, 36.1°N , 120.3°E). To ensure the reliability of the experimental results, a blank PVC board was selected as the control group, and different observation periods were chosen based on the seasons. The samples were extracted at regular intervals within the specified time range, and the growth of fouling organisms was recorded to evaluate the antifouling ability of the resin coatings. Furthermore, all the samples were immediately placed back into the seawater after capturing photographs to prevent the organisms from dying because of long-term exposure to air.

4. Results and discussion

4.1. Structural characterization

4.1.1. ASM synthesis

The chemical structures of HEMA, ASM, and RSN-6018 were analyzed via ATR-IR spectroscopy. Fig. 2(a) depicts that for the ASM, the Si–OH peak at 3379 cm^{-1} is not observed, whereas the characteristic peak of C=O appears at 1722 cm^{-1} , indicating the successful reaction between HEMA and RSN-6018 [8,39]. Furthermore, considering the thermal analysis results (Figs. 2(b) and (c)), because $E_{\text{C-C}}$ (347 kJ/mol) < $E_{\text{C-O}}$ (360 kJ/mol) < $E_{\text{C-H}}$ (413 kJ/mol) < $E_{\text{Si-O}}$ (452 kJ/mol), RSN-6018 modified with HEMA underwent an additional –C–O– bond-breaking process compared with RSN-6018. The aforementioned findings revealed that HEMA successfully modified RSN-6018 for the preparation of the ASM.

4.1.2. Synthesis of $\text{ASM}_x\text{-Zn}$

The analysis of a series of prepolymers and their corresponding zinc resins using ATR-IR and ^1H NMR spectroscopic techniques provides several crucial insights. First, Fig. 3(a) exhibits the peaks of a benzene ring and Si–O–C at 750 and 500 cm^{-1} , respectively. Additionally, from Fig. 3(d), the characteristic peaks of the hydrogen atoms originating from the phenyl component in the RSN-6018 material appear at $\delta = 7.2\text{--}8.5 \text{ ppm}$. These characteristics confirm the successful copolymerization of the ASM with acrylate monomers. Furthermore, in Fig. 3(b), a distinct characteristic peak [40] is observed at 1608 cm^{-1} , which is attributed to the $\text{Zn}(\text{COO})_2$ group that is commonly referred to as zinc carboxylate. Considering the ^1H NMR results depicted in Figs. 3(c) and

(d), the prepolymer (pre- ASM_x) spectrum exhibits a characteristic peak [41] of –COOH at $\delta = 12.3 \text{ ppm}$. However, in the $\text{ASM}_x\text{-Zn}$ spectrum, no such characteristic peak is detected at the aforementioned position, indicating the successful reaction of –COOH with ZnO. In summary, the abovementioned findings confirm the successful preparation of organosilicon-modified zinc acrylate polymers.

4.1.3. XPS measurements of $\text{ASM}_x\text{-Zn}$ coatings

To determine the elemental composition of the polymer coating after curing, XPS characterization was performed on the $\text{ASM}_{0.5}\text{-Zn}$ polymer coating. The results are presented in Fig. 4. From Figs. 4(a) and (b), the zinc content on the coating surface is 0.31 % [42]. This indicates that after the curing of the polymer, zinc ions are distributed on the coating surface as hydrolyzed functional groups, facilitating the self-polishing effect of the coating in marine environments. From Figs. 4(a) and (c), the silicon content on the coating surface is 12.54 %. This suggests that the polysiloxane structure is located on the surface of the coating rather than inside, which considerably enhances the adhesion and hydrophobicity of the coating, enabling a self-cleaning effect in marine environments.

4.2. Hydrolysis properties

Self-polishing coatings must exhibit a consistent polishing rate in practical applications. The polishing of the self-polishing antifouling coatings is achieved using acrylic polymers containing numerous easily hydrolyzed side chains. The $\text{ASM}_x\text{-Zn}$ polymer coating was immersed in real seawater for 35 d for weight-loss testing. From Fig. 5(a), the $\text{ASM}_x\text{-Zn}$ polymer coating hydrolyzes within 35 d, producing a large amount of free COO^- on the surface [8]. However, the mass loss of the coating is large at the early stage of hydrolysis and gradually stabilizes at the later stages. This may be attributed to the high content of ester groups and zinc ions in the polymer side chain at the early stage, resulting in the rapid hydrolysis of the coating. However, at the later stages, the decrease in the content of ester groups and zinc ions reduces the rate of hydrolysis of the coating and maintains a stable polishing rate to achieve a stable release of zinc ions, inhibiting the growth of fouling organisms. From Fig. 5(b), the weight loss of the ASM fluctuates with the change in the ASM content over 35 d, and the mass losses of only $\text{ASM}_{0.5}\text{-Zn}$ and $\text{ASM}_0\text{-Zn}$ are consistent. The higher mass loss is considered to be because of the presence of a polysiloxane structure with a high molecular weight in the ASM, and the excess ASM will hinder the copolymerization of acrylate monomers, resulting in a lower yield (Table S2). This leads to a large amount of free ASM within the polymer, which is released during the hydrolysis of the coating, increasing the weight loss of $\text{ASM}_{1.5}\text{-Zn}$ to 16.4 %. However, the smaller mass loss of $\text{ASM}_{0.25}\text{-Zn}$ may be because of the lower ASM content in the polymer, which leads to a smaller number of ester groups in the polymer, decreasing the hydrolysis rate of the coating and resulting in a smaller weight loss. In summary, the silicone-

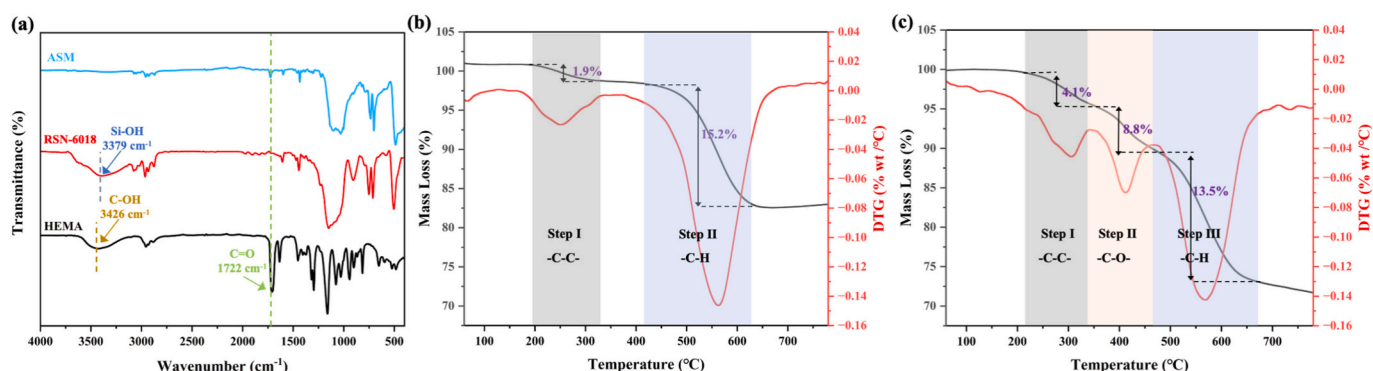


Fig. 2. Material characterization of the ASM: (a) IR spectrum of the ASM. TGA results for (b) RSN-6018 and (c) ASM.

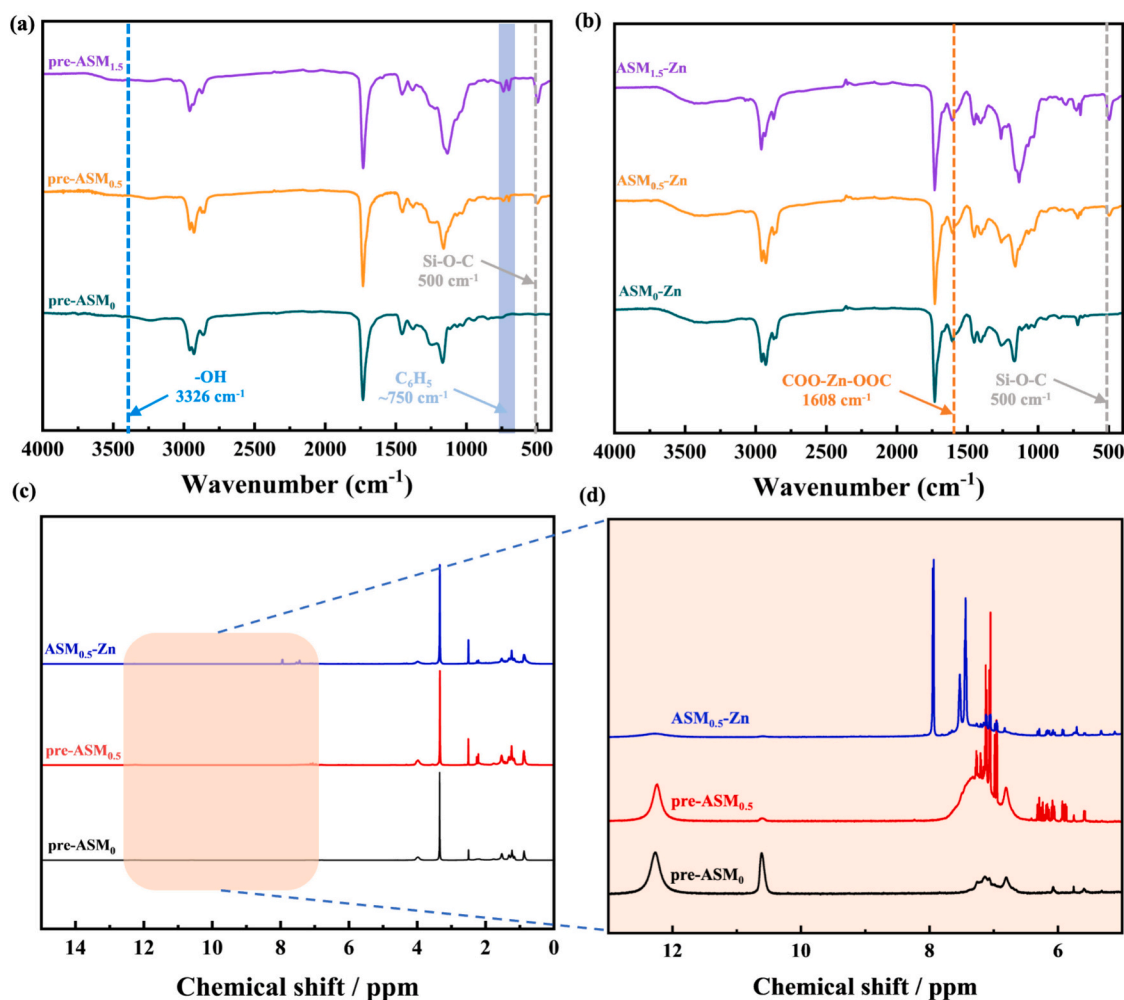


Fig. 3. IR spectra of (a) pre- ASM_x and (b) $\text{ASM}_x\text{-Zn}$ polymer. (c) ^1H NMR spectra and (d) enlargement of the parts of the spectra of pre- ASM_x and $\text{ASM}_x\text{-Zn}$ polymer.

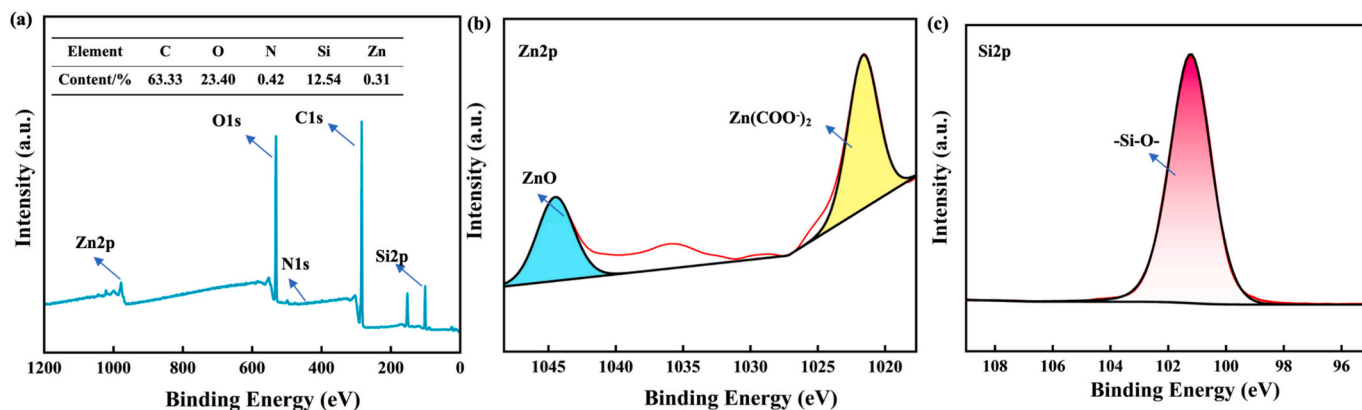


Fig. 4. Surface element analysis of the $\text{ASM}_{0.5}\text{-Zn}$ polymer coating. (a) XPS full spectrum of the $\text{ASM}_{0.5}\text{-Zn}$ polymer. Distribution maps for (b) zinc and (c) silicon elements in the $\text{ASM}_{0.5}\text{-Zn}$ polymer.

modified acrylic coatings exhibit good self-polishing properties.

4.3. Wettability of $\text{ASM}_x\text{-Zn}$ coatings

The Si-O bond in polysiloxane structures exhibits a long bond length ($1.65\text{ \AA}$), a large bond angle (159°), and a substantially high bond energy (452 kJ/mol). These properties enable the polymer to form a film after the side end group at the interface of close stacking, demonstrating

good conformational mobility. Meanwhile, the main chain of silicon atoms linked to the alkane side group can be shielded using the high-energy silicon-oxygen chain. Furthermore, due to the relatively low polarity of the C-H bond, the intermolecular interactions between the silicone main chains are significantly weakened. Consequently, these reasons lead to a reduction in surface free energy, causing the coating to exhibit pronounced hydrophobicity. [43]. From Fig. 6(a), the addition of small and excessive amounts of the ASM may alter the hydrophilicity of

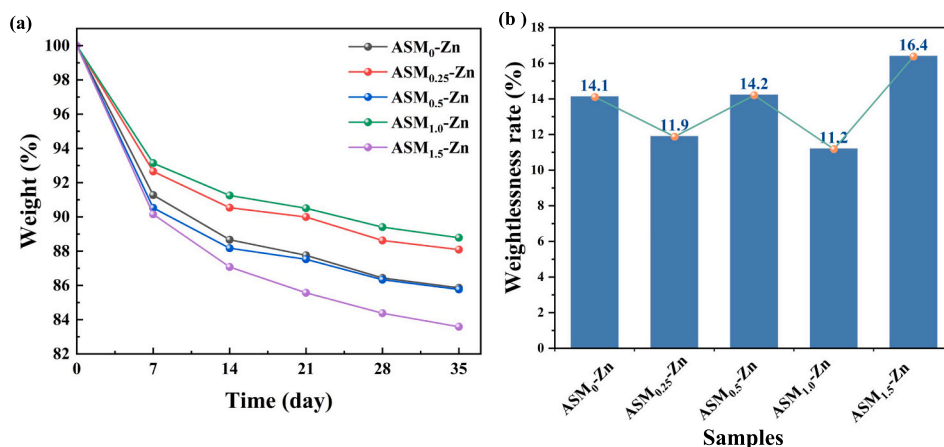


Fig. 5. (a) Weight-loss ratios of ASM_x-Zn coatings. (b) Weightlessness rate of ASM_x-Zn coatings soaked in seawater for 35 d.

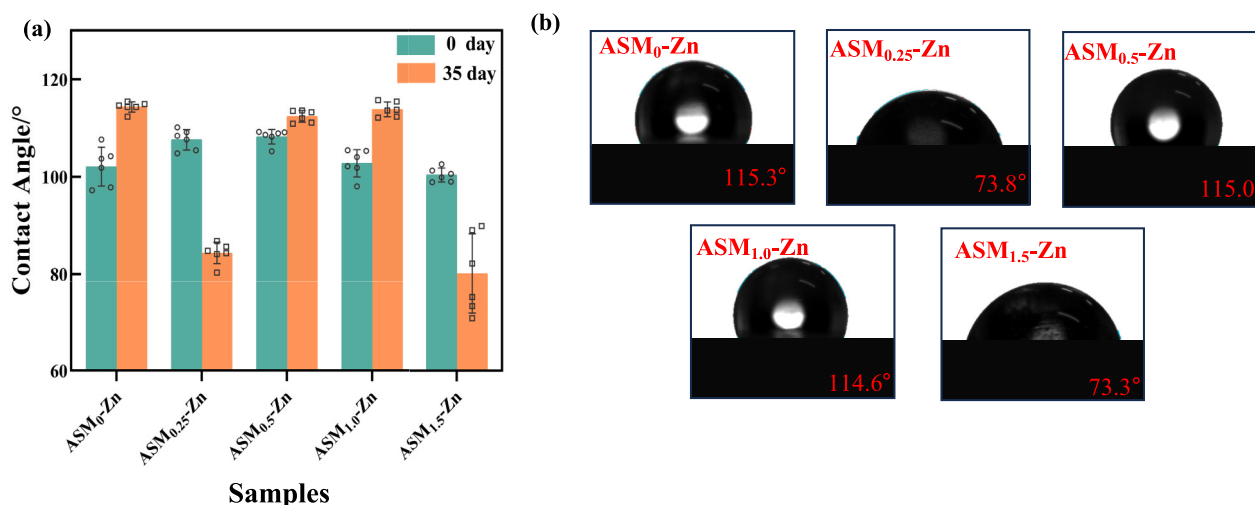


Fig. 6. (a) Contact angles (CA) of ASM_x-Zn coatings before and after immersion in seawater for 35 d. (b) Images depicting the contact angles of ASM_x-Zn coatings after soaking in seawater for 35 d.

the coatings. Combining these results with the lower ASM_{0.25}-Zn and ASM_{1.5}-Zn polymer conversions in Table S2, fewer polysiloxane structures and n-dodecyl groups are present on the polymerized surface of the coatings, which leads to a decrease in hydrophobicity. The contact angles of ASM_{0.5}-Zn and ASM_{1.0}-Zn remain at approximately 115° after 35 d, and the higher content of polysiloxane structures and n-dodecyl groups polymerized in these two proportions impart stable hydrophobicity to the coatings, which provide the basis for the detachment of marine fouling organisms from the coated surfaces.

4.4. Surface morphology of ASM_x-Zn coatings

Because fouling organisms generally have difficulty to adhere to smooth surfaces, the surface roughness of coatings plays a crucial role in preventing the adhesion of fouling organisms [44]. The effects of the ASM on the surface roughness of the polymer coatings were quantitatively evaluated using laser confocal microscopy. The ASM_{0.25}-Zn polymer coating exhibits a maximum roughness of 1.841 μm after 14 d of immersion in seawater. The roughness values of ASM₀-Zn, ASM_{0.5}-Zn, ASM_{1.0}-Zn, and ASM_{1.5}-Zn coatings are approximately 0.696, 0.866, 0.887, and 0.524 μm, respectively, indicating relatively smooth surfaces. The laser confocal topography analysis reveals that the addition of the ASM with $x = 0.25$ results in a dense phase separation after the immersion of coating in seawater. This may be caused by the hydrolysis

of the coating surface [45], leading to the generation of pores in the coating and numerous free hydrophilic COO⁻ on the surface, which is nonuniform and comprises hydrophobic polysiloxane structures [46]. Consequently, the *Ra* of the coating increases. With the increase in the ASM content, the *Ra* of the coating gradually decreases. The hydrolysis of the hydrophobic n-dodecyl group leads to the migration of polysiloxane structures to the surface. This migration reduces the degree of phase separation on the coating surface, resulting in a smoother finish and a lower *Ra* value (Fig. 7).

4.5. Transparency of ASM_x-Zn coatings

Several coatings that exhibit antifouling functions require the use of transparent plates [47]. When a text is placed behind a coating that has been soaked in real seawater for 7 d, it remains readable, except for the ASM_{1.5}-Zn polymer coating (Fig. 8(a)). From Figs. 8(b) and (c), the transmittance of the coatings increases with increasing wavelength, and the transmittance of the ASM_{1.0}-Zn polymer coating at the wavelengths of >450 nm is >80 %, exhibiting the highest transmittance at 800 nm. After immersion in seawater, the transmittance of ASM_{1.5}-Zn is 0 % at all the tested wavelengths, which is consistent with the results in Fig. 8 (a). This is because of the hydrolysis of the coating surface or the release of polysiloxane structures in large numbers, resulting in the entry of water inside the coating. Fig. 8(d) presents a comparison of the transmittance

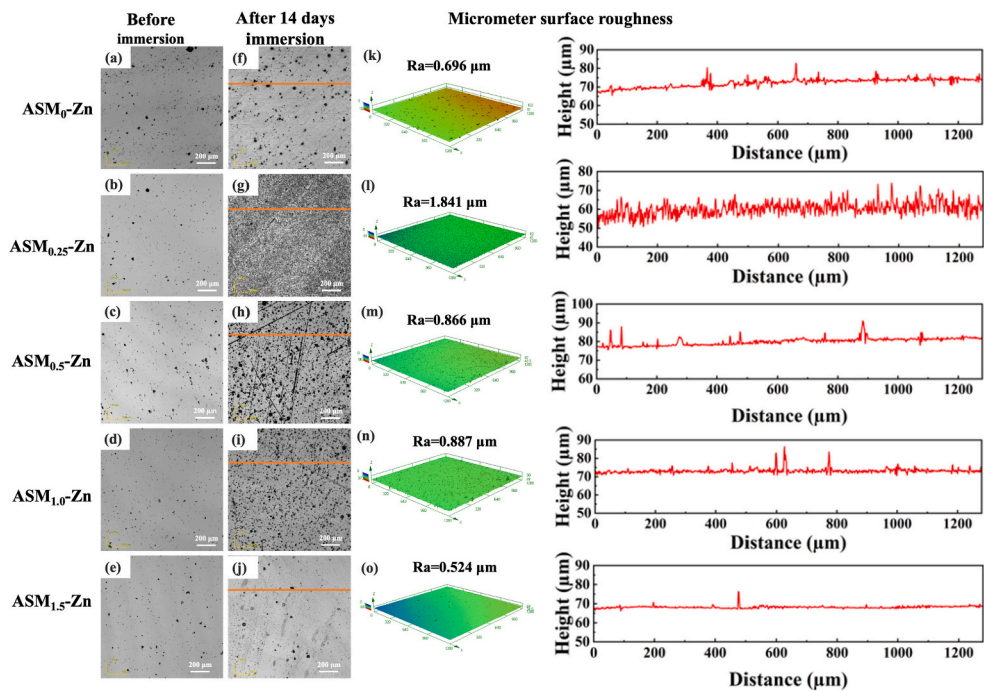


Fig. 7. Representative laser confocal images of ASM_x -Zn coatings (a–e) before immersion and (f–j) after 14 d of immersion in real seawater. (k–o) Representative three-dimensional images and roughness (R_a) of ASM_x -Zn coatings.

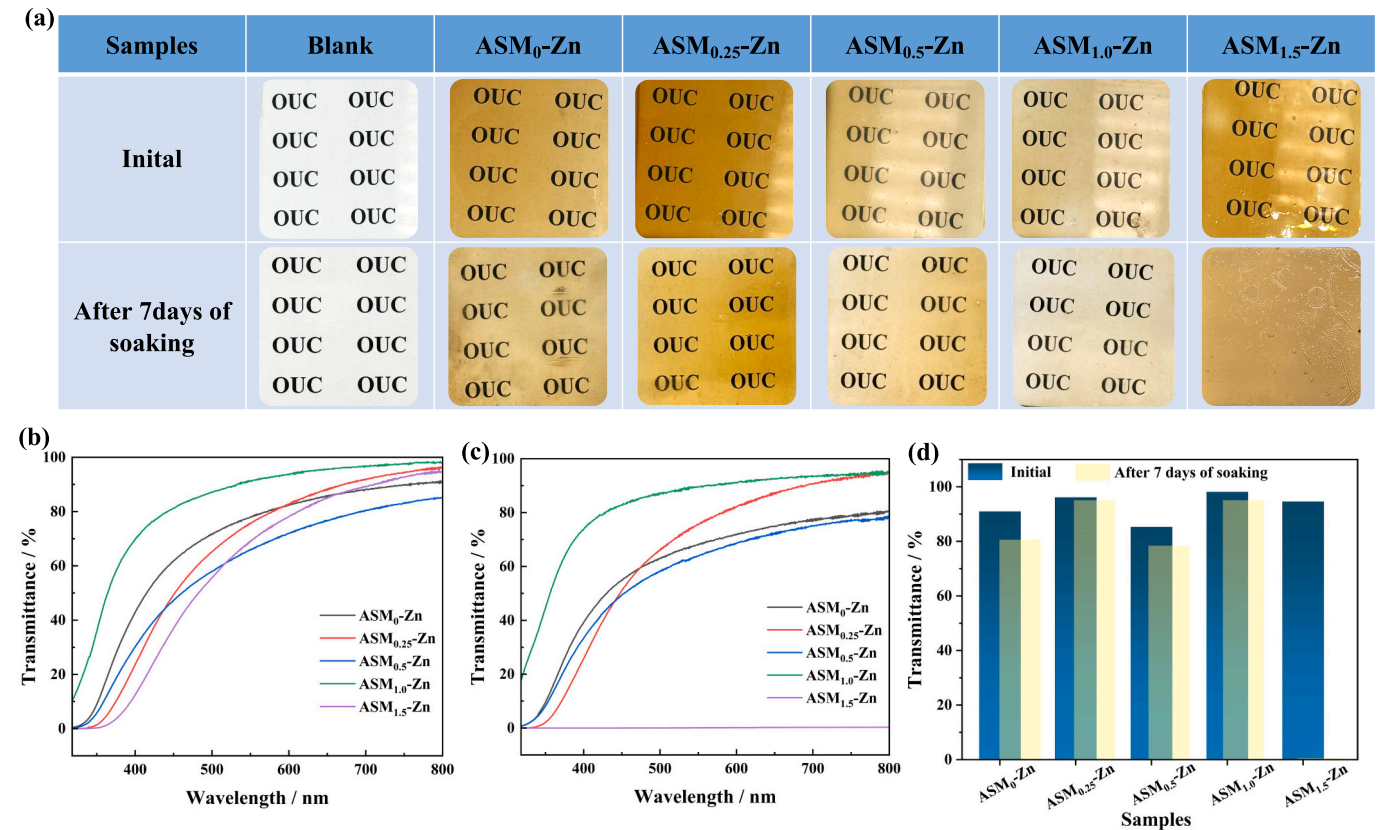


Fig. 8. Transparency tests of ASM_x -Zn copolymer coatings. (a) Transparency of ASM_x -Zn copolymer coatings tested using the given text before and after immersion in real seawater for 7 d. UV-vis transmittance spectra of ASM_x -Zn copolymer coatings (b) before and (c) after immersion in real seawater for 7 d. (d) Comparison of the transmittance values at 800 nm of ASM_x -Zn copolymer coatings before and after immersion in real seawater for 7 d.

values at 800 nm of the coatings before and after immersion in seawater. The transmittance of the coatings decreases after immersion in seawater, which may be related to the hydrolysis of the ester group of the coatings. This exposes hydrophilic COO^- , causing water molecules to enter the coating, which reduces the transmittance of the coatings.

4.6. Analysis of the mechanical properties of the coatings

The mechanical properties of silicone-modified acrylic polymer coatings are very important, as they determine the duration for which the coating can remain intact in the real-sea environment. Therefore, the silicone-modified acrylic polymer coatings were subjected to adhesion, bending, and pencil hardness tests. Fig. 9(a) depicts the effects of different ASM contents on the interlayer adhesion of the coating layers. The interlayer adhesion of the $\text{ASM}_{0.5}\text{-Zn}$ coating was the strongest, exhibiting a value of 1.25 MPa. Fig. 9(b) presents the effects of different ASM contents on the flexibility of the coatings. $\text{ASM}_{0.5}\text{-Zn}$ and $\text{ASM}_{1.0}\text{-Zn}$ coatings exhibited better flexibility than the other analyzed coatings, and the surfaces of these coatings remained flat after 50 bending cycles. Fig. 9(c) illustrates the effects of different ASM contents on the hardness of the coatings, and the hardness values of $\text{ASM}_{0.5}\text{-Zn}$ and $\text{ASM}_{1.0}\text{-Zn}$ coatings are lower than those of the other analyzed coatings. These results indicated that the coating surfaces were softer, and the as-prepared coatings could be used on devices with irregular surfaces. Considering the aforementioned test results, the polymer coatings were placed in a dynamic paddling experimental device to evaluate their seawater

scouring resistance. Fig. 9(d) depicts that when the as-prepared coatings are rotated at 80 rpm, only the $\text{ASM}_{0.5}\text{-Zn}$ coating is maintained flat, which indicates that the silicone-modified acrylic polymer coatings can maintain good mechanical properties under seawater conditions.

4.7. Antibacterial assay

The unique Si-O-Si structure of polysiloxane exhibits excellent hydrophobicity, functioning as a natural barrier that effectively reduces the contact between the coating surface and moisture and nutrients. Furthermore, it inhibits bacterial adhesion and proliferation and severs the chain of bacterial biofilm formation through surface attachment. Additionally, the coating surface is adorned with zinc ions and organic acids, which, in synergy with the hydrophobic properties, offers a formidable antibacterial defense, imposing a potent restraint on the growth of harmful bacterial species, such as *E. coli* and *S. aureus*.

A meticulous examination of Fig. 10(a) reveals that the antibacterial efficacy of the coating is optimal when the ASM content is 0.5. This is because the ASM, at this concentration, serves as a catalyst to facilitate more efficient hydrolysis reactions on the coating surface (Fig. 5(b)). Furthermore, the polysiloxane structure enhances the hydrophobicity of the coating, maximizing its antibacterial performance.

Figs. 10(b) and (c) further emphasize the exceptional inhibitory effects of $\text{ASM}_x\text{-Zn}$ coatings against gram-positive bacteria. These effects originate from the high permeability of the cell walls and membranes of the gram-positive bacteria to zinc ions, enabling these ions to effortlessly

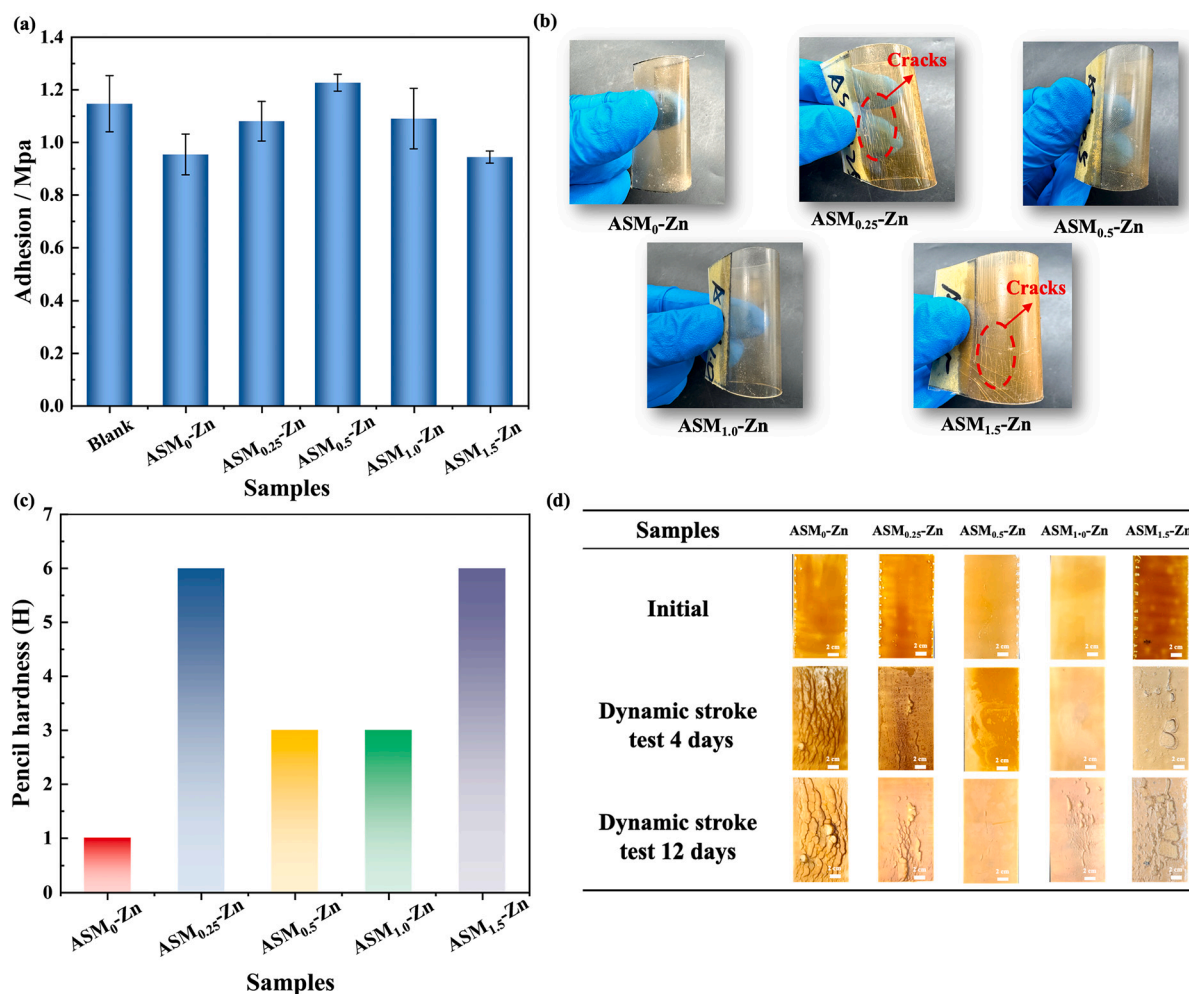


Fig. 9. Mechanical property analysis of $\text{ASM}_x\text{-Zn}$ copolymer coatings. (a) Adhesion, (b) flexibility, and (c) pencil hardness of the coatings before immersion in seawater. (d) Flatness of the coatings before and after water erosion under dynamic conditions.

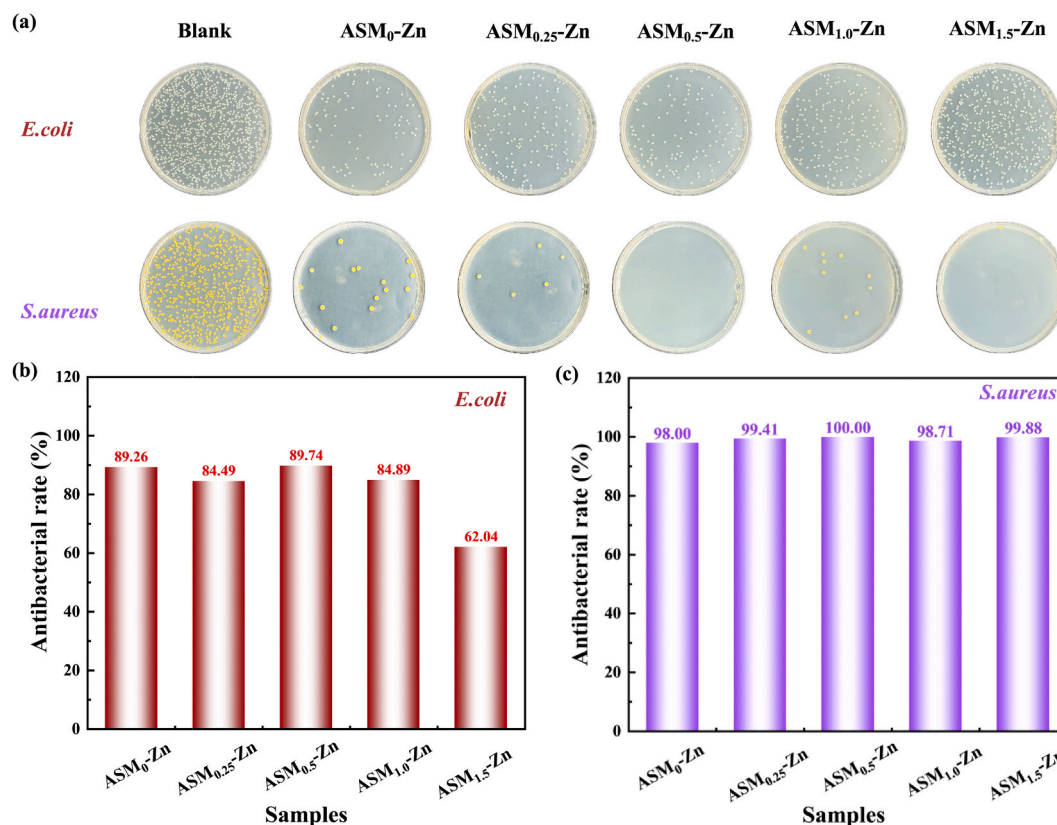


Fig. 10. Effects of ASM_x-Zn copolymer coatings on *E. coli* and *S. aureus*. (a) Comparison of bacterial growth. Inhibition rates of (b) *E. coli* and (c) *S. aureus*.

penetrate the bacterial interior. When the zinc ions reach the interior of bacterial cells, they disrupt the integrity of the cell walls and membranes, interfere with normal metabolic processes, and inhibit enzymatic activities. Consequently, under identical zinc-ion concentrations, gram-positive bacteria are subjected to a more pronounced inhibitory effect than the gram-negative bacteria. In summary, ASM_x-Zn coatings possess considerable potential for application in marine environments to effectively prevent the growth of primary biofilms and inhibit the attachment and proliferation of fouling organisms, providing a highly reliable protective covering for marine engineering facilities.

4.8. Algal inhibition experiment

Employing ASM_x-Zn polymer coatings, we conducted separate algae inhibition experiments on green algae (*C. pyrenoidosa*) and diatoms (*N. closterium*). Polysiloxane structures are renowned for their excellent antiadhesion properties against algae [48,49]. Fig. 11 illustrates that after the addition of the ASM to the coating, the inhibitory effect of the initial coating on *N. closterium* is considerably retained; however, the efficacy of the coating against *C. pyrenoidosa* unexpectedly decreases. These observations may be attributed to the distinctive biological traits of green algae (*C. pyrenoidosa*): their thin cell walls, rapid growth, and high reproductive capacity enhance their adhesion and growth on coatings under the current experimental conditions. Additionally, the incorporation of the ASM may modify the physical properties of the copolymer, particularly by potentially decreasing its weight-loss rate. These alterations further curtail the release of zinc ions and organic acids during the hydrolysis of the copolymer. As zinc ions and organic acids are pivotal algae-inhibiting components that typically disrupt the Ca²⁺ balance [40] within algal cells, which contribute to the inhibitory effects of the coatings, a decrease in their concentrations correspondingly weakens their effects on the growth of green algae. This provides a plausible explanation for the notable decrease in the inhibition rate of

the coatings against green algae (*C. pyrenoidosa*) after the introduction of the ASM.

4.9. Marine field tests

ASM_x-Zn coatings have comprehensively exhibited exceptional ability to inhibit bacteria and algae in controlled laboratory settings as well as demonstrated excellent stability in marine environments. As depicted in Fig. 12, a rigorous two-month field test of the antifouling properties of the coatings was conducted from July 19th to September 19th, 2024, in Qingdao City, Shandong Province, located in the Yellow Sea of China (coordinates: 36.1°N, 120.3°E). This testing period coincided with the peak growth season of fouling organisms, presenting a formidable natural challenge for the verification of the efficacy of the coatings. The zinc acrylic polymer coating devoid of the ASM initially displayed a limited fouling-release effect, owing to its long-chain alkyl groups. However, within 60 d of exposure to the harsh marine environment, it underwent considerable deterioration because of intense seawater erosion. Consequently, its surface was rapidly overtaken by fouling organisms, such as barnacles, leading to a substantial decrease in its antifouling performance. Conversely, the acrylic zinc copolymer coating modified with organosilicon exhibited an excellent performance in the marine environment. This coating effectively addressed the pervasive issue of insufficient interlayer adhesion in organosilicon structures and demonstrated exceptional antifouling capabilities in practical applications. Furthermore, the modified coating resisted the erosive forces of seawater and effectively prevented the adhesion of fouling organisms, establishing a robust defense for the sustained and safe operation of marine facilities. Furthermore, this groundbreaking research provides valuable inspiration and guidance for the development of acrylic copolymer coatings that simultaneously possess self-polishing and hydrophobic properties.

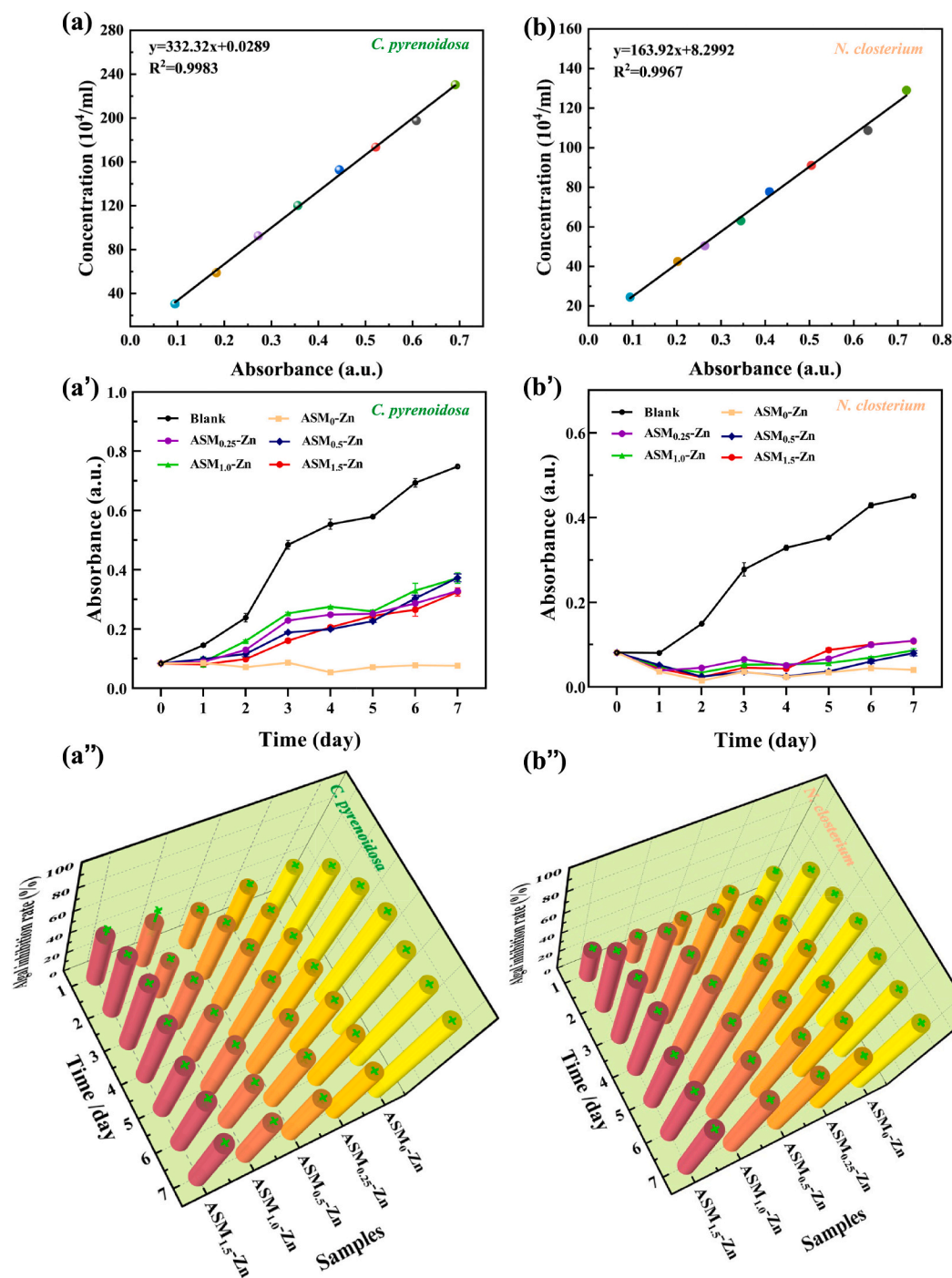


Fig. 11. Effects of ASM_x-Zn copolymer coatings on *Chlorella pyrenoidosa* (C. pyrenoidosa) and *Nitzschia closterium* (N. closterium). Standard curves for (a) C. pyrenoidosa and (b) N. closterium. Growth inhibition curves of ASM_x-Zn copolymer coatings for (a') C. pyrenoidosa and (b') N. closterium. Algal inhibition rates of ASM_x-Zn copolymer coatings for (a'') C. pyrenoidosa and (b'') N. closterium.

5. Conclusions

This study ingeniously combines two advantageous materials, namely, silicone and zinc acrylate resin, to innovatively prepare transparent and homogeneous silicone-modified zinc acrylate copolymer (ASM_x-Zn) coating. As a cost-effective hydrophobic material, silicone endows the coatings with excellent hydrophobic properties (CA = 115.0°). Secondly, zinc acrylate resin, regarded as the key element of self-polishing antifouling coatings, ensures that the coatings possess a stable self-polishing rate during 35 days of seawater immersion. In addition, the ASM component can precisely regulate the transmittance

of the coatings in seawater environments, providing strong technical support for the antifouling requirements of marine optical devices. At the same time, the coatings exhibit excellent mechanical properties, maintaining exceptional adhesion and flexibility even under harsh conditions, such as seawater erosion, ultimately ensuring the smoothness and cleanliness of the coating surfaces. Notably, the zinc ions released by ASM_x-Zn coatings exhibit excellent antibacterial properties which enhances the antifouling capability of the coatings and provides robust support for their long-term application in marine environments. In actual tests conducted in sea areas, the antifouling effects of the prepared coatings lasted for more than two months, successfully

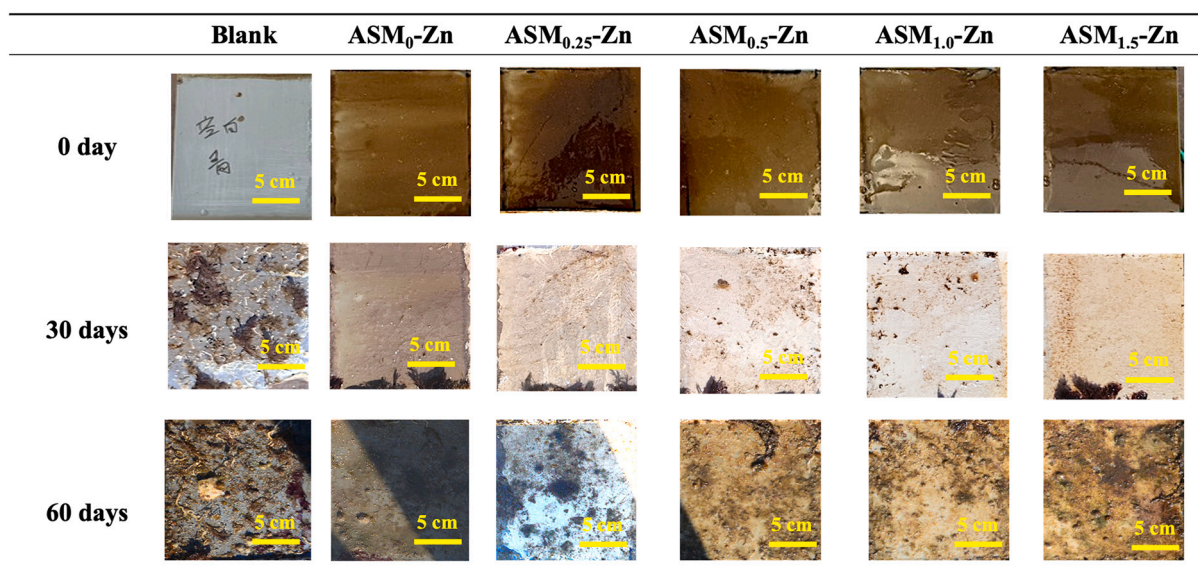


Fig. 12. Marine antifouling performance test of ASM_x -Zn coatings in the Qingdao Yellow Sea area. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

confirming their reliability and stability in practical applications. The as-developed coatings are expected to play an important role in future applications in the field of marine engineering.

CRediT authorship contribution statement

Zheng Wang: Methodology, Investigation, Writing – review & editing, Writing – original draft. **Enhui He:** Data curation, Writing – review & editing. **Guanglei Song:** Visualization. **Xiaohui Jiang:** Project administration. **Liangmin Yu:** Resources, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.porgcoat.2025.109143>.

Data availability

The authors do not have permission to share data.

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